Improving-Workplace-Safety-Using-Data.R

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```r
# Load the packages
library("readr")
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library("ggplot2")
library("broom")

# Import the data
hr_data <- read_csv("~/Desktop/Data/hr_data_2.csv")
```

```
## Rows: 2940 Columns: 4

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): location
## dbl (3): year, employee_id, overtime_hours
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
accident_data <- read_csv("~/Desktop/Data/accident_data.csv")
```

```
## Rows: 302 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): accident_type
## dbl (2): year, employee_id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Create hr_joined with left_join() and mutate()
hr_joined <- left_join(hr_data,accident_data, by = c("year","employee_id")) %>%
  mutate(had_accident = ifelse(is.na(accident_type), 0, 1))

hr_joined
```

```
## # A tibble: 2,940 x 6
##     year employee_id location    overtime_hours accident_type had_accident
##    <dbl>       <dbl> <chr>                <dbl> <chr>                <dbl>
##  1  2016           1 Northwood               14 <NA>                     0
##  2  2017           1 Northwood                8 Mild                     1
##  3  2016           2 East Valley              8 <NA>                     0
##  4  2017           2 East Valley             11 <NA>                     0
##  5  2016           4 East Valley              4 <NA>                     0
##  6  2017           4 East Valley              2 Mild                     1
##  7  2016           5 West River               0 <NA>                     0
##  8  2017           5 West River              17 <NA>                     0
##  9  2016           7 West River              21 <NA>                     0
## 10  2017           7 West River               9 <NA>                     0
## # i 2,930 more rows
```

```r
# Find accident rate for each year
hr_joined %>%
  group_by(year) %>%
  summarize(accident_rate = mean(had_accident))
```

```
## # A tibble: 2 x 2
##    year accident_rate
##   <dbl>         <dbl>
## 1  2016        0.0850
## 2  2017        0.120
```

```r
# Test difference in accident rate between years
chisq.test(hr_joined$year, hr_joined$had_accident)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  hr_joined$year and hr_joined$had_accident
## X-squared = 9.5986, df = 1, p-value = 0.001947
```

```r
# Which location had the highest acccident rate?
hr_joined %>%
  group_by(location) %>%
  summarize(accident_rate = mean(had_accident)) %>%
  arrange(desc(accident_rate))
```

```
## # A tibble: 4 x 2
##   location    accident_rate
##   <chr>               <dbl>
## 1 East Valley         0.128
```

```
## 2 Southfield         0.103
## 3 West River         0.0961
## 4 Northwood          0.0831
```

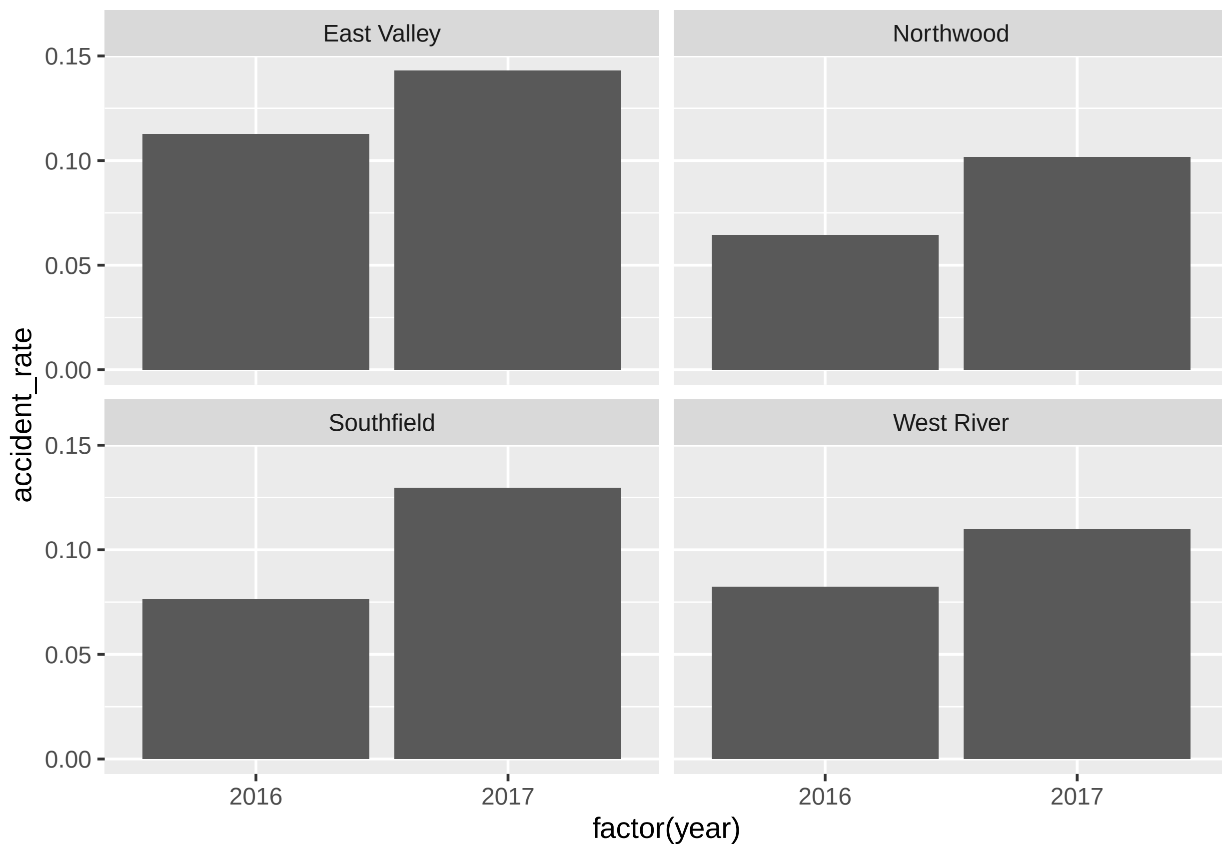
```
# Compare annual accident rates by location
accident_rates <- hr_joined %>%
  group_by(location, year) %>%
  summarize(accident_rate = mean(had_accident))
```

```
## `summarise()` has grouped output by 'location'. You can override using the
## `.groups` argument.
```

```
accident_rates
```

```
## # A tibble: 8 x 3
## # Groups:   location [4]
##   location     year accident_rate
##   <chr>       <dbl>         <dbl>
## 1 East Valley  2016        0.113
## 2 East Valley  2017        0.143
## 3 Northwood    2016        0.0644
## 4 Northwood    2017        0.102
## 5 Southfield   2016        0.0764
## 6 Southfield   2017        0.130
## 7 West River   2016        0.0824
## 8 West River   2017        0.110
```

```
# Graph it
accident_rates %>%
  ggplot(aes(factor(year), accident_rate)) +
  geom_col() +
  facet_wrap(~location)
```




```r
# Filter out the other locations
southfield <- hr_joined %>%
  filter(location == "Southfield")


# Find the average overtime hours worked by year
southfield %>%
  group_by(year) %>%
  summarize(average_overtime_hours = mean(overtime_hours))
```

```
## # A tibble: 2 x 2
##    year average_overtime_hours
##   <dbl>                  <dbl>
## 1  2016                   8.67
## 2  2017                   9.97
```

```r
# Test difference in Southfield's overtime hours between years
t.test(overtime_hours ~ year, data = southfield)
```

```
##
##  Welch Two Sample t-test
##
## data:  overtime_hours by year
## t = -1.6043, df = 595.46, p-value = 0.1092
## alternative hypothesis: true difference in means between group 2016 and group 2017 is not equal to 0
```

```
## 95 percent confidence interval:
##  -2.904043  0.292747
## sample estimates:
## mean in group 2016 mean in group 2017
##           8.667774           9.973422
```

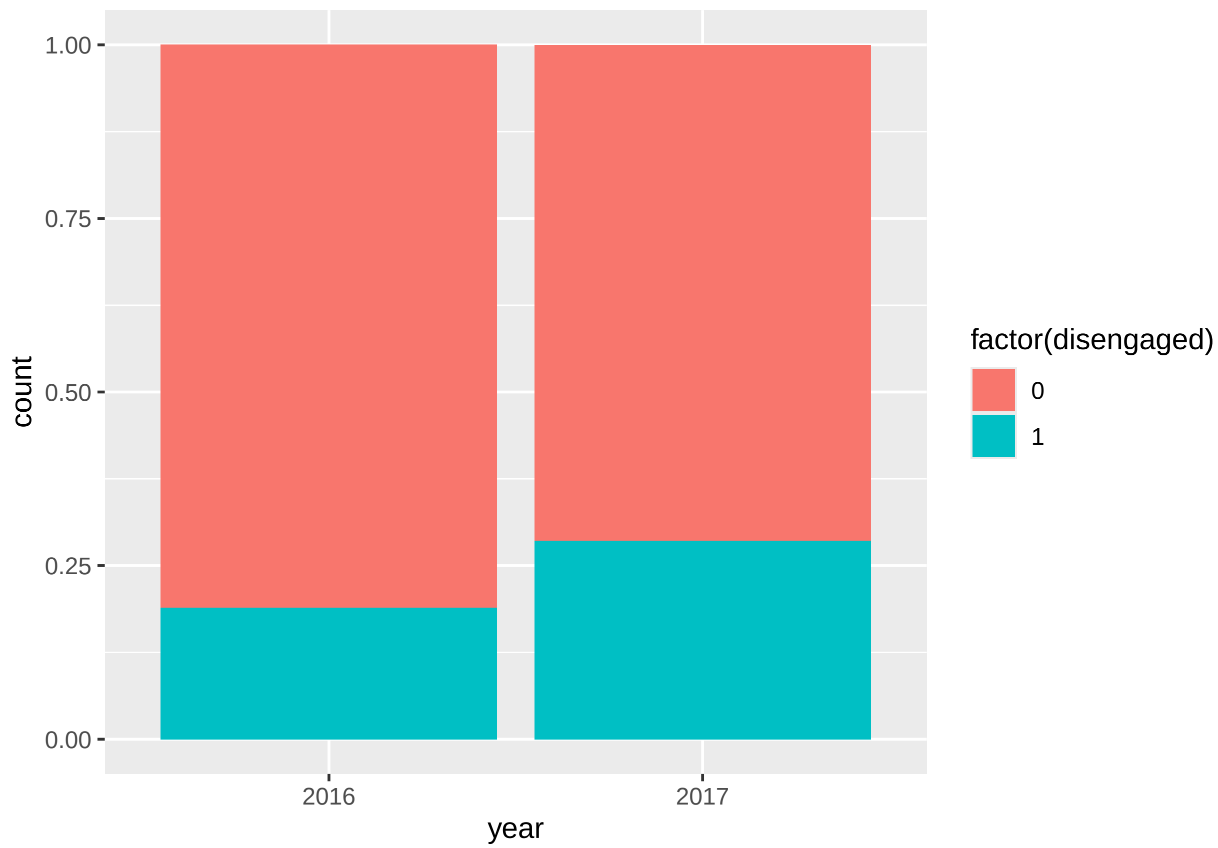
```r
# Import the survey data
survey_data <- read_csv("~/Desktop/Data/survey_data_2.csv")
```

```
## Rows: 2940 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (3): year, employee_id, engagement
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Create the safety dataset
safety1 <- left_join(hr_joined, survey_data, by = c("year", "employee_id")) %>%
  mutate(disengaged = ifelse(engagement <= 2, 1, 0),
         year = factor(year)) %>%
  filter(location == "Southfield")

safety <- left_join(hr_joined, survey_data, by = c("year", "employee_id")) %>%
  mutate(disengaged = ifelse(engagement <= 2, 1, 0),
         year = factor(year))

# Visualize the difference in % disengaged by year in Southfield
safety1 %>%
  ggplot(aes(x = year, fill = factor(disengaged))) +
  geom_bar(position = "fill")
```




```
# Test whether one year had significantly more disengaged employees
chisq.test(safety1$year, safety1$disengaged)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  safety1$year and safety1$disengaged
## X-squared = 7.1906, df = 1, p-value = 0.007329
```

```
# Filter out Southfield
other_locs <- safety %>%
  filter(location != "Southfield")
head(other_locs)
```

```
## # A tibble: 6 x 8
##   year  employee_id location     overtime_hours accident_type had_accident
##   <fct>       <dbl> <chr>                 <dbl> <chr>                <dbl>
## 1 2016            1 Northwood                14 <NA>                     0
## 2 2017            1 Northwood                 8 Mild                     1
## 3 2016            2 East Valley               8 <NA>                     0
## 4 2017            2 East Valley              11 <NA>                     0
## 5 2016            4 East Valley               4 <NA>                     0
## 6 2017            4 East Valley               2 Mild                     1
## # i 2 more variables: engagement <dbl>, disengaged <dbl>
```

```r
# Test whether one year had significantly more overtime hours worked
t.test(overtime_hours ~ year, data = other_locs)
```

```
##
##  Welch Two Sample t-test
##
## data:  overtime_hours by year
## t = -0.48267, df = 2320.3, p-value = 0.6294
## alternative hypothesis: true difference in means between group 2016 and group 2017 is not equal to 0
## 95 percent confidence interval:
##  -0.9961022  0.6026035
## sample estimates:
## mean in group 2016 mean in group 2017
##           9.278015           9.474765
```

```r
# Test whether one year had significantly more disengaged employees
chisq.test(other_locs$year, other_locs$disengaged)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  other_locs$year and other_locs$disengaged
## X-squared = 0.0023091, df = 1, p-value = 0.9617
```

```r
# Use multiple regression to test the impact of year and disengaged on accident rate in Southfield
regression <- glm(had_accident ~ year + disengaged, family = "binomial", data = safety1 )

# Examine the output
tidy(regression)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -2.92      0.250    -11.7  1.74e-31
## 2 year2017       0.440     0.285      1.55 1.22e- 1
## 3 disengaged     1.44      0.278      5.19 2.13e- 7
```